

University of Texas at El Paso (UTEP) Institutional Review Board
Informed Consent Form for Research Involving Human Subjects

Protocol Title: Measurements of decision-making rationality

Principal Investigator: Alexander Friedman

UTEP: Biology

Introduction

You are being asked to take part voluntarily in the research project described below. You are encouraged to take your time in making your decision. It is important that you read the information that describes the study. Please ask the study researcher or the study staff to explain any words or information that you do not clearly understand.

Why is this study being done?

Approximately 400 adult individuals, 18 years and older will be enrolling in this study at UTEP. We will record physiological variables like heart rate, heart rate variability, respiration rate, pupil dilation, and eye movement.

To record the heart rate, heart rate variability, blood pressure and respiration rate we will use Wahoo heart rate monitor, wearing this equipment is fully elective for each participant. To measure the eye movement and pupil dilation we will use Tobii Pro Spark eye tracker. The eye tracker camera captures images of both eyes for accurate measurement of eye gaze and eye position in 3D space, as well as pupil diameter it is well integrated between the eye tracker time domain and the client computer time domain and provides an accurate calculation of the data and a good level of precision. The eye tracker works by sending out near infrared light (IR) and the light is reflected in the subjects' eyes. And those reflections are picked up by the eye tracker's camera. And through filtering and calculations, the eye tracker knows where subjects are looking. We will conduct this behavioral experiment through an already developed app which is actually a questionnaire presented to the participants in the form of stories. And each participant will be asked to complete 2-8 stories per session, and the total number of completed sessions can vary by participant availability and which contexts the participant can complete up to a maximum of 21 sessions possible; this flexibility is meant to fill underpowered contexts and strengthen statistical validity. On day 1 the participants will go through some basic questions like

age range, sex, if experiencing any pain, level of tiredness and then they will be able to pick topics/stories of their choice from the list of stories available, and each story will have 6 preference questions and 16 decision-making questions. From day 2 to the rest of the sessions, participants will go through the same format of decision-making questions. These app sessions will be conducted before and after lunch hours and/or early and late hours of the day. During the session the eye tracker will be on to collect eye movement data and subjects *who voluntarily elect to*, will be wearing the heart rate monitor. Time and day for each session will be determined based on the subjects schedule.

We will chart subjects' ability to perform cognitive and moral decision-making at a strategic level. We will identify the imbalanced decision-making observed in healthy human subjects such as overvaluation of rewards, cost, or risk in decision-making. We will use a combination of neuronal network, algorithmics and analytic models to classify the psychometric functions in our subjects. To identify the patterns of striosomal circuit activation that led to optimal cost-benefit decision-making, we will develop a predictive model.

What is involved in the study?

If you agree to take part in this study, the research team will collect data from a sample size of a total of 400. You are being asked to be in the study because we will chart subjects' ability to perform cognitive and moral decision-making at a strategic level. We will record physiological variables like heart rate, heart rate variability, respiration rate, pupil dilation, and eye movement (cardiac metrics are collected for participants who voluntarily opt into wearing the heart rate monitor). We will conduct this behavioral part of the experiment through an already developed app which is a questionnaire presented to the participants in the form of stories and each participant will be asked to perform up to 21 sessions. On day 1 the participants will go through some basic questions like age range, sex, if experiencing any pain, level of tiredness and then they will be able to pick topics/stories of their choice from the list of stories available, and each story will have 6 preference questions and 16 decision-making questions. From day 2 till the end of the sessions, participants will go through the same format decision-making questions. These app sessions will be conducted before and after lunch hours and/or early and late hours of the day. After those sessions by using computational modeling, we will identify the

imbalanced decision-making observed in healthy human subjects such as overvaluation of rewards, cost, or risk in decision-making.

If you decide to enroll in this study, your involvement will be asked to complete up to a max of 21 sessions. Each session is about 20-30 minutes. Each session has 3-4 stories consisting of 6 choice preference questions and 16 decision-making questions. During the session the eye tracker will be on to collect eye movement data and you will be asked whether you would elect to wear a heart rate monitor, if accepted the heart rate monitor is worn as a sport strap around the chest and if declined the experimental process does not change except for progressing without wearing the strap (The project has a proposed duration of three years). Time and day for each session will be determined based on the subjects schedule.

What are the risks and discomforts of the study?

The risks associated with this research are expected to be no greater than those involved in daily activities.

Due to the nature of the study, even what are thought to be solely benign, 'surface-level' questions for minor psychological discomfort due to unknown adverse responses to certain questions or scenario content. Further, there is minimal risk of loss of confidentiality though the five-digit ID number and data de-identification processes are in place to reduce this possibility.

What will happen if I am injured in this study?

The University of Texas at El Paso and its affiliates do not offer to pay for or cover the cost of medical treatment for research related illness or injury. No funds have been set aside to pay or reimburse you in the event of such injury or illness. You will not give up any of your legal rights by signing this consent form. You should report any such injury to *Dr. Alexander Friedman*, 857-233-1843 and to the UTEP Institutional Review Board (IRB) at (915-747-6590) or irb@utep.edu.

Are there benefits to taking part in this study?

You are not likely to benefit by taking part in this study. (The successful implementation of our research plan will enable the development of new tools for assessing brain, manipulating state and physiological state in the real world; and the development of new tools to classify types of decision-makers in real time. It will be transformative to conceptual understanding of decision-

making, which will create a new training capacity within the U.S. military; the ability to measure optimal decision-making in healthy officers and service members. We will create new brain circuit-based cognitive neuro-biofeedback therapies that will promote strategic thinking in officer, whose decisions determine the future of our democracy in crisis situations. Our enhanced understanding of the brain circuits that support decision-making may have implication for semi-automated decision-making by supervised artificial intelligence systems.)

What are my costs?

There are no direct costs.

Will I be paid to participate in this study?

You will not be compensated for taking part in this research study.

What other options are there?

You have the option not to take part in this study. There will be no penalties involved if you choose not to take part in this study.

Choosing to withdraw or not participate will not affect your grades, class, or university standing.

Choosing to withdraw or not participate will not affect your employment, departmental, or university standing.

What if I want to withdraw, or am asked to withdraw from this study?

Taking part in this study is voluntary. You have the right to choose not to take part in this study. If you do not take part in the study, there will be no penalty or loss of benefit.

If you choose to take part, you have the right to skip any questions or stop at any time. However, we encourage you to talk to a member of the research group so that they know why you are leaving the study. If there are any new findings during the study that may affect whether you want to continue to take part, you will be told about them.

The researcher may decide to stop your participation without your permission, if he or she thinks that being in the study may cause you harm.

Who do I call if I have questions or problems?

You may ask any questions you have now. If you have questions later, you may call Dr. Friedman at 857-233-1843 and afriedman@utep.edu

If you have questions or concerns about your participation as a research subject, please contact the UTEP Institutional Review Board (IRB) at (915-747-6590 or irb@utep.edu).

What about confidentiality?

Your part in this study is confidential. The following procedures will be followed to keep their personal information confidential.

The results of this research study may be presented at meetings or in publications; however, your name will not be disclosed in those presentations.

Every effort will be made to keep your information confidential. Your personal information may be disclosed if required by law.

Organizations that may inspect and/or copy your research records for quality assurance and data analysis include, but are not necessarily limited to:

- Office of Human Research Protections
- UTEP Institutional Review Board

Because of the need to release information to these parties, absolute confidentiality cannot be guaranteed.

All research involved will keep detailed electronic experimental data stored in a database. These notes will have sequentially marked pages, where any experiment will be documented in detail. All notes and protocols during experiments will be collected by our software. The database will be managed by MATLAB and Proc SQL functions. All notes will be recorded and updated daily. It will be stored on the UTEP server. Experimental setup computers and data on the server will be backed up weekly. We will

create a detailed protocol description for all functions and data as routinely done for our past publications. We will use read-me files to create a supplemental material document with a description of all the functions and procedures. Data generated from this project will be administered in accordance with both University and NSF policies, including NSF data sharing policy and implementation guidance. Friedman Laboratory follows an open data philosophy. Therefore, in our previous publications and this proposal, all data and protocols without exclusion will be deposited into the Mendeley Database. All codes used to analyze and process the data will be submitted to GitHub database. We will create special scripts that generate published figures and panels from raw data and include statistical analysis. Any inventions developed under this project will be reported and managed as provided by NSF policies. The electronic version of the final, upon acceptance for publication will be deposited to the National Library of Medicine's Pubmed Central to be made publicly available no later than one year after the official date of publication. Both the PI and the University of Texas at El Paso will have responsibilities, and hence rights, concerning access to, use of, and maintenance of original research data.

Mandatory reporting

N/A

Authorization Statement

I have read each page of this paper about the study (or it was read to me). I will be given a copy of the form to keep. I know I can stop being in this study without penalty. I know that being in this study is voluntary and I choose to be in this study.

Participant's Name (printed)

Participant's Signature

Date

Signature of Person Obtaining Consent

Date